Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

RAJSHAHI-6204, BANGLADESH

A CASE STUDY REPORT

ON

A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing

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DEPARTMENT OF INDUSTRIAL & PRODUCTION ENGINEERING

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This is to certify that the case study entitled "A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing" is carried out by Riasad Rahman Rakib, Roll No.: 2105019, and Hasin Almas Sanim, Roll No.: 2105060, under my supervision in the Department of Industrial & Production Engineering of Rajshahi University of Engineering & Technology.

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CANDIDATES' DECLARATION

It is hereby declared that this case study or any part of it has not been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

As cities in developing countries keep expanding, handling all that trash gets a lot harder especially when it comes to actually collecting and hauling it away. These two jobs alone eat up most of the budget. In this paper, we lay out a smarter, two-step approach to tackle the messiest problems: complicated collection routes and how to get the most out of the truck fleet. Our focus is Rajshahi City Corporation (RCC) in Bangladesh. At first, we dug into ideas from the Clustered Vehicle Routing Problem (CluVRP) and some solid Operations Research. We put together a model that works in two stages. In the first stage, we use unsupervised machine learning in plain language, K-Means clustering to split up the city's Secondary Transfer Stations (STS) into clear, non-overlapping zones for collection. Each station gets its own turf. After that, we bring in a Mixed Integer Linear Programming (MILP) model to iron out the details. This piece figures out the most efficient way to send out a bunch of different trucks, not just picking how many, but also considering things like their different sizes, how many trips each one can make, and the strict 8-hour shifts for workers. Our main goal: drive down daily costs fuel, labor, even fines from carbon emissions. In the end, we look at what all this means for city planning, and how a system like this can really help clean up urban supply chains and make them more efficient.

Keywords: Municipal Solid Waste Management (MSWM), Mixed Integer Linear Programming (MILP), K-Means Clustering, Clustered Vehicle Routing Problem (CluVRP), Route Optimization, Sustainable City Logistics, Developing Urban Economies, Heterogeneous Fleet Management

TABLE OF CONTENTS

Title **Page No.**

ACKNOWLEDGEMENT iv

ABSTRACT v

TABLE OF CONTENTS vi

LIST OF TABLES viii

LIST OF FIGURES ix

NOMENCLATURE x

CHAPTER 1 12

INTRODUCTION 12

1.1 Introduction 12

1.2 Problem Statement 13

1.3 Objectives 13

1.4 Organization of the Case Study 14

CHAPTER 2 15

LITERATURE REVIEW 15

2.1 Review of Literature 15

CHAPTER 3 21

METHODOLOGY 21

3.1 Mathematical Model 22

3.1.1 Sets and Indices 23

3.1.2 Parameters 23

3.1.3 Decision Variables 24

3.1.4 Objective Function 24

3.1.5 Constraints 25
3.2 Assumptions 28
3.3 Solution Approach 28
CHAPTER 4 31
RESULT AND DISCUSSION 31
4.1 Data Collection 31
4.2 Result Analysis 33
4.2.1 Experiment of K-means clustering algorithm 33
4.2.2 Experiment of the proposed MILP model 35
4.3 Sensitivity Analysis 37
4.3.1 Impact of Carbon Tax (λ) 37
4.3.2 Impact of Fuel Price 38
4.3.3 Impact of Traffic Congestion (β) 40
4.3.4 Combined Effect of Fuel Price and Traffic Congestion 41
4.4 Discussion 42
4.4.1 Model Performance 42
4.4.2 Sensitivity to Urban Dynamics 42
4.4.3 Environmental Policy Impact 43
4.4.4 Combined Stress Effects 43
4.4.5 Practical Implications 43
CHAPTER 5 44
CONCLUSION AND RECOMMENDATIONS 44
5.1 Conclusion 44
5.2 Recommendations 44
REFERENCES 46

LIST OF TABLES

Table No.	Title	Page No.
Table 2.1	An overview of the review of literature	17
Table 4.1	Waste collecting vehicle information	32
Table 4.2	Parameters involved	33
Table 4.3	STS names with their coordinates and waste generations of each STS per day	34
Table 4.4	Cost Reduction Analysis	35
Table 4.5	Impact of carbon tax	37
Table 4.6	Impact of fuel price	39
Table 4.7	Impact of traffic congestion	40

LIST OF FIGURES

Fig. No.	Title	Page No.
Fig. 3.1	Vehicle Routing in a single cluster	21
Fig. 3.2	Flowchart of the overall process	29
Fig. 4.1	Clustering of STS	34
Fig. 4.2	Comparative economic analysis: Traditional vs Optimal Collection	36
Fig. 4.3	Impact of carbon tax on daily cost	38
Fig. 4.4	Impact of fuel price on daily cost	39
Fig. 4.5	Impact of traffic congestion on daily cost	40
Fig. 4.6	Heatmap of traffic vs fuel price on daily cost	41

NOMENCLATURE

STS : Secondary Transfer Station

MILP : Mixed Integer Linear Programming

RCC : Rajshahi City Corporation

MSWM : Municipal Solid Waste Management

VRP : Vehicle Routing Problem

GIS : Geographic Information System

CVRP : Capacitated Vehicle Routing Problem

CluVRP : Clustered Vehicle Routing Problem

CVRPTW : Capacitated Vehicle Routing Problem with Time Windows

CHAPTER 1

INTRODUCTION

1.1 Introduction

Municipal Solid Waste Management is an intricate logistical challenge, deeply tied to urban sustainability and public health. In developing cities like Rajshahi, Bangladesh, the collection and transport of waste account for up to 70-80% of total MSWM expenditures. Recent empirical data from RCC indicates a daily waste generation dominated by organic material (approximately 70%), which necessitates prompt removal to prevent severe environmental degradation [1].

Despite the urgency, existing collection networks often rely on heuristic human dispatching, leading to overlapping routes, underutilized vehicle capacities, and significant redundant mileage. As urban sprawl continues, the reliance on manual routing fails to scale, exacerbating fuel costs and greenhouse gas emissions. While theoretical literature frequently addresses the Vehicle Routing Problem (VRP) and its variants, translating these mathematical equations into actionable, solver-ready models for city planners remains difficult. Specifically, models often fail to account for the rigid realities of labor laws (e.g., strict 8-hour shifts) and the necessity for vehicles to execute multiple trips dynamically without relying on continuous variable creation during solver execution. This paper seeks to formalize an optimization framework tailored for RCC that bridges the gap between theoretical CluVRP formulations [1] and computational solver realities. A cluster-first route-second constructive heuristic based on continuous approximation (CA) has been proposed to efficiently dispatch relief commodities such as fuel in post-disaster scenarios, demonstrating improved routing efficiency under emergency conditions[2]

1.2 Problem Statement

The collection and transportation of municipal solid waste in Rajshahi City Corporation (RCC) currently consume up to 80% of the total waste management budget[1]. This inefficiency stems from heuristic, manually planned routing that results in overlapping service zones, redundant mileage, and sub-optimal fleet utilization. While mathematical optimization models like the Vehicle Routing Problem (VRP) offer theoretical solutions[3], they frequently fail in practical municipal applications because they are computationally intractable for city-scale networks and ignore rigid real-world constraints such as strict 8-hour labor shifts and the necessity for multi-trip landfill cycles. Therefore, a critical need exists for a computationally efficient, localized optimization framework that can partition the city into manageable clusters and generate cost-effective, constraint-compliant routing schedules for a heterogeneous transport fleet.

1.3 Objectives

1. To develop an integrated two-stage framework that utilizes K-Means clustering for geometric partitioning of secondary transfer station (STS) networks to improve spatial efficiency.
2. To formulate a Mixed Integer Linear Programming (MILP) model for optimizing waste collection routing using a heterogeneous fleet consisting of direct shuttle vehicles and multi-stop sweepers.
3. To design a multi-objective optimization function that minimizes total system cost, including fuel consumption, vehicle operation (wage) cost, and carbon emissions.
4. To ensure operational feasibility by incorporating key real-world constraints such as capacity limits, time windows, flow balance, and shift restrictions within the routing model.
5. To evaluate the robustness and scalability of the proposed model under real-world economic and environmental variability through extensive parametric sensitivity analysis.

1.4 Organization of the Case Study

Chapter 2: The Literature Review examines existing research on the Municipal Solid Waste Management System in the urban cities.

Chapter 3: This methodology chapter presents the overall research framework.

Chapter 4: The results and discussion chapter present the prioritization outcomes.

Chapter 5: The conclusion chapter summarizes the major findings and future scopes of the study.

CHAPTER 2

LITERATURE REVIEW

2.1 Review of Literature

Municipal solid waste management (MSWM) has received increasing attention due to its critical role in ensuring environmental sustainability, public health, and urban livability. The complexity of MSWM systems arises from the interaction of waste generation, collection, transportation, treatment, and disposal processes, all of which must be managed under economic, environmental, and operational constraints.

Solid Waste Management in Rajshahi City

A number of studies have looked into how Rajshahi City Corporation (RCC) is currently handling its municipal solid waste. Habib et al. [1] calculated that the city produces about 358 tons of municipal solid waste every day, which is about 0.42 kg per person per day. Their research showed that a lot of trash is still not picked up or is thrown away in open dumps because the collection coverage is limited and the transportation systems are not very good. These actions cause pollution, greenhouse gas emissions, and health risks for the public.

Das et al.[4] studied waste composition and energy recovery potential by utilizing landfill gas in Rajshahi. Their study emphasized that the methane emissions are positively related to irregular waste collection and uncontrolled dumping, thereby calling for systematic and optimized waste collection approaches. However, their research was almost entirely about environmental impacts and did not include operational improvements related to routing or transportation planning.

Afrin et al. [5] presented a GIS-based method to enhance the waste collection point efficiency by implementing shortest-path techniques for routing. Although their study reported possible reductions in the total distance travelled as well as collection time, it did not address vehicle capacity constraints, multi-trip routing or integral integration with higher-level system

optimization. Furthermore, collection point clustering has not been included as a decision variable to facilitate its scalability and realism.

In summary, the literature based on Rajshahi City mostly focuses on the waste generation characteristics and descriptive level analyses rather than integrated optimization frameworks simultaneously considering clustering, routing and operating constraints.

Optimization Models for Solid Waste Management Systems

Various researchers have proposed mathematical optimization models for MSWM systems, particularly using mixed-integer linear programming (MILP), to increase the efficiency of MSWM systems [6], [7]. Shaban et al. [7] presented a holistic MILP model for the design of a sustainable municipal solid waste management system, which considers waste sources, transfer stations, recycling facilities and disposal sites. Their model identifies best facility locations, waste flows and system configurations that minimize total cost.

While this framework gives important information to infrastructure level planning, it assumes transportation in an aggregate way without having specific model for the vehicle routing decisions. Consequently, operational characteristics including route connectivity, vehicle capacity utilization, and travel time requirements are not accounted for in their entirety.

Other studies based on optimization also emphasize facility location and waste flow assignment [7], [8], [9] highlighting the success of using MILP in strategic planning. However, these models typically assume deterministic waste generation and overlook uncertainties in demand, which can significantly affect system feasibility in real-world scenarios.

Vehicle Routing and Hybrid Collection Systems

Vehicle routing problems (VRPs) have been extensively used to optimize waste collection and transport. Hannan et al.[8] proposed a route optimization model using cost and emission objectives, concluding that optimized routing can substantially cut fuel and environmental impact. Sitek et al.[10] addresses a highly complex variant of vehicle routing problem, specifically capacities vehicle routing problem with time window. Though their approach does rely on real-time data, but has no integration of routing decisions with clustering or system-level planning.

People have looked into dynamic routing, too. For example, Sreelekshmi and Nair [11] designed a dynamic vehicle routing method that uses clustering and metaheuristic algorithms to keep up with changing demand. That's a great way to add flexibility, but most of these methods just assume all vehicles are the same. They miss how real waste collection usually works.

In reality, waste collection relies on mixed fleets. Big trucks or shuttles trucks carry large loads between main sites, while smaller vehicles, or sweepers, focus on picking up trash in neighborhoods. Even though this setup is common and makes a lot of sense, not many studies actually dig into it, especially when it comes to optimization models.

Clustering-Based Routing and Integrated MILP Approaches

Clustering plays a big role in making routing problems more manageable. By splitting service areas into smaller regions, teams can tackle each section more efficiently. Alesiani et al. [12] introduced a constrained clustering method for capacitated vehicle routing problems, and it sped up computation. Battarra and colleagues [13] delved into cluster-based routing, pointing out that keeping routes compact matters a lot in city services.

Still, most methods stick to only clustering or routing routine. They set up clusters or just dive into route optimization. Those are straight forward processes, but misses out on the advantages of optimizing clustering and routing together.

With recent MILP modeling techniques, integrating clustering directly into the routing problem is now possible. Yet, most current models don't bring together clustering decisions, routing constraints, and broader system goals especially when it comes to municipal solid waste management.

Table 2.1: An overview of the review of literature

Author	Year	Model	Title	Findings	Limitations
Md. A. Habib et al. [14]	2021	Empirical analysis + waste characterization	Municipal Solid Waste Management and Waste-to-Energy Potential from Rajshahi City Corporation	Identified that Rajshahi generates ~358 tons/day of waste with inefficient collection systems leading to open dumping and environmental hazards. Highlighted potential for energy recovery.	Focused on waste characteristics and energy potential; lacks optimization of collection and routing systems.
B. K. Das et al. [4]	2014	Environmental assessment + landfill gas modeling	Energy Recovery Potential from Municipal Solid Waste in Rajshahi City by Landfill Technique	Demonstrated that improper waste collection leads to methane emissions; landfill gas can be used	Does not address routing, clustering, or operational optimization of waste collection.

				for energy generation.	
S. Afrin et al. [5]	2023	GIS-based shortest path model	A GIS-Based Solid Waste Management of Rajshahi City Corporation	Showed that optimized shortest-path routing reduces travel distance and improves collection efficiency. Identified lack of bins and poor routing.	Ignores vehicle capacity, multi-trip routing, clustering, and system-level optimization.
A. Shaban et al. [7]	2022	Mixed-Integer Linear Programming (MILP)	An Optimization Model for the Design of a Sustainable Municipal Solid Waste Management System	Developed MILP model for optimal facility location, waste flow, and system design minimizing cost.	Transportation modeled in aggregated form; routing, clustering, and operational constraints not explicitly included.
M. A. Hannan et al. [8]	2020	MILP-based routing optimization	Waste Collection Route Optimisation Model for Linking Cost Saving and Emission Reduction	Demonstrated that optimized routing reduces fuel cost and carbon emissions significantly.	Requires real-time data; does not integrate clustering or hybrid vehicle systems.
S. V. Sreelekshmi & J. Nair [11]	2017	Dynamic VRP + clustering (K-means + VNS)	Dynamic Vehicle Routing for Solid Waste Management	Showed clustering improves routing efficiency and adaptability under dynamic conditions.	Assumes homogeneous vehicles; lacks integration with system-level planning and MILP formulation.
F. Alesiani et al. [12]	2022	Constrained clustering for CVRP	Constrained Clustering for the	Demonstrated that clustering reduces	Not applied specifically to waste

			Capacitated Vehicle Routing Problem	computational complexity and improves solution efficiency.	management; clustering not integrated with full system optimization.
M. Battarra et al. [13]	2014	Clustered Vehicle Routing Problem (CluVRP)	Exact Algorithms for the Clustered Vehicle Routing Problem	Showed that clustering improves route compactness and operational feasibility in routing problems.	Computationally expensive; not integrated with waste management system constraints.
Sitek et al.[10]	2021	Capacited Vehicle Routing Problem with Time Window	Optimization of capacitated vehicle routing problem with alternative delivery, pick-up and time windows: A modified hybrid approach	The inclusion of time windows and logical constraints increases computational complexity across all solution methods, while strict enforcement of time windows leads to a higher number of routes and vehicles required to serve the same demand.	The objective function minimizes distance and penalties but ignores key factors such as total time, trips, and resource use, limiting its ability to fully reflect real-world operational efficiency.

Table 2.1 presents a summary of the existing literature on municipal solid waste management and vehicle routing optimization. It highlights the methodologies, key findings, and limitations of prior studies, emphasizing the research gap in integrated clustering and routing approaches under real-world operational constraints.

CHAPTER 3

METHODOLOGY

The study proposes a two-tier, data-driven optimization framework that integrates unsupervised machine learning with exact mathematical formulation to solve a specialized Capacitated Vehicle Routing Problem with Time Windows (CVRPTW). The algorithmic workflow is executed in two consecutive phases to manage computational complexity and ensure logical fleet deployment.

Spatial Partitioning via K-Means Clustering

The RCC service area contains a set of Secondary Transfer Stations (STS) where primary waste is temporarily deposited. To eliminate overlapping routes, the unsupervised K-Means algorithm [15] deployed to partition the STSs into K distinct clusters. The algorithm minimizes the within-cluster sum of squares, ensuring that geographically proximate STSs are grouped [16] . Each cluster represents an isolated routing zone served by specific vehicles, which drastically reduces the dimensionality of the subsequent routing problem [6].

Phase 1 & 2: Route Optimization via MILP

Once the clusters are defined, a MILP model is executed for each cluster. Instead of allowing the solver to generate trips “on the fly” (which violated the rigid matrix requirements of computational solvers), an upper bound for the set of required trips M based on the total waste volume of the cluster and the maximum capacity of the assigned fleet. The model then optimally assigns STS nodes to specific trips and sequence.

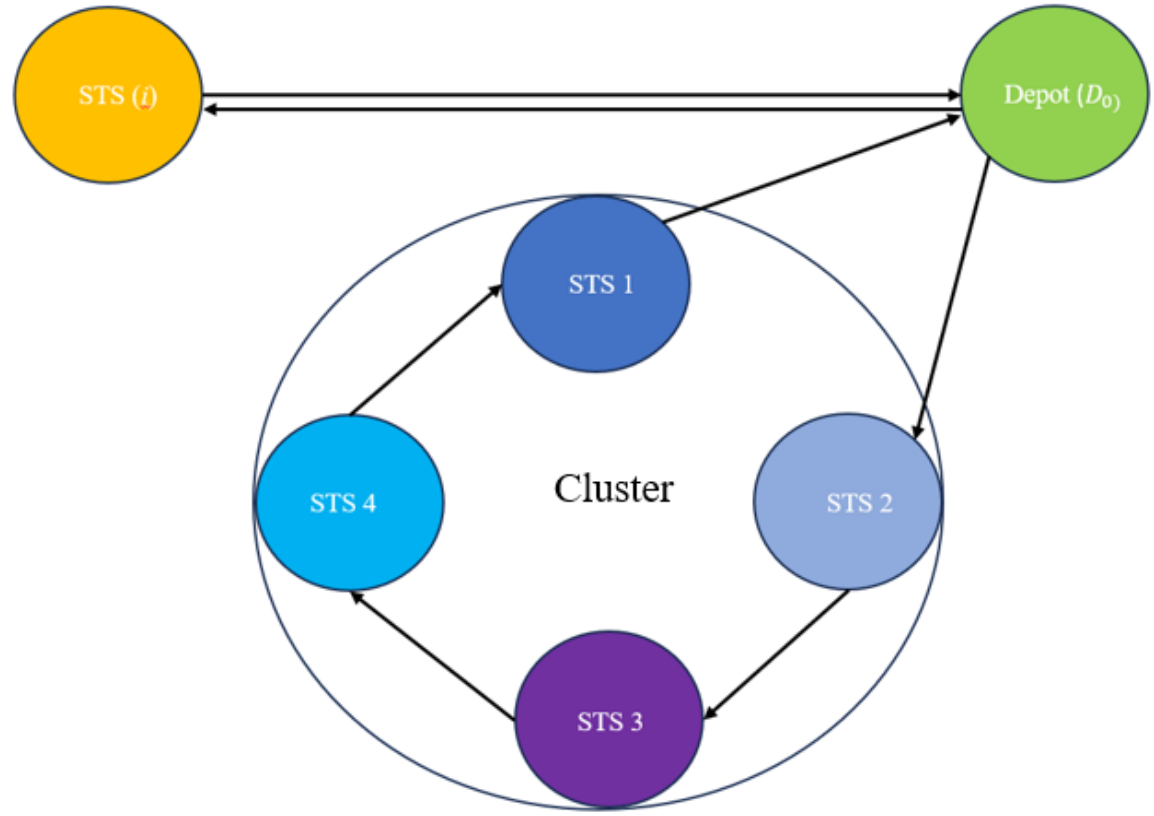


Figure 3.1: Vehicle Routing in a single cluster

Figure 3.1 illustrates the routing structure within a single cluster, where multiple secondary transfer stations (STS) are grouped spatially. A sweeper vehicle starts from the depot, visits clustered STS nodes in a sequential route, and returns to the depot, ensuring efficient waste collection within the cluster.

3.1 Mathematical Model

The mathematical model for the proposed system with corresponding set and indices, parameters, decision variables, objective functions, and constraints are given here.

3.1.1 Sets and Indices

- $i, j \in \mathcal{N}$: Set of all network nodes, where node 0 denotes the central Depot and $\mathcal{C} = \mathcal{N} \backslash \{0\}$ is the set of Secondary Transfer Station (STS) nodes within a given cluster.
- $k \in K_S$: Set of Shuttle vehicles, responsible for primary bulk waste extraction via direct depot-to-STS round-trips.
- $k \in K_R$: Set of Sweeper vehicles, responsible for secondary fractional waste collection along multi-stop tours.
- $m \in M$: Set of trip slots; $m \in \{1,2, \dots, |M|\}$, where $|M|$ is the maximum number of sequential tours a sweeper may complete within a single operational shift.
- $\mathcal{A}$: Set of admissible arcs, defined as

$$\mathcal{A} = \left\{ (i,j) \;\middle|\; \frac{2\, D_{0i}}{v/\beta} + t_{LU}^{R} \;\leq\; T_{\max}, \quad i \neq j \right\}.$$

3.1.2 Parameters

- Q_k: Payload capacity of vehicle k (tonnes).
- C_k^{fuel}: Per-kilometre fuel cost of vehicle k (BDT/km).
- C_k^{wage}: Fixed daily labour cost of vehicle k (BDT/day).
- E_k: CO_2 emission factor of vehicle k (kg CO_2/km).
- λ: Carbon penalty rate (BDT/kg CO_2).
- $\overline{S}_i$: Nominal waste accumulation at STS i (tonnes).
- $\hat{S}_i$: Maximum surge demand above the nominal level at STS i (tonnes).
- Γ: Robustness budget; $\Gamma \in [0,1]$.
- D_{0i}: Road-network distance from the Depot to STS i (km).
- D_{ij}: Road-network distance from node i to node j (km).
- v: Free-flow vehicle travel speed (km/h).

- β: Urban traffic congestion multiplier ($\beta \geq 1$); effective travel speed is v/β.
- t_{LU}: Loading and unloading service time per stop (hours).
- $[E_i, L_i]$: Hard time window at STS i: earliest permitted arrival E_i and latest permitted arrival L_i (hours).
- B_{dur}: Mandatory driver rest duration per deployed vehicle (hours).
- $T_{\max}$: Maximum permissible shift duration (hours).
- $\mathcal{M}$: Dynamic big-M constant for time-continuity linearisation (hours).

3.1.3 Decision Variables

- $y_{ik} \in \mathbb{Z}_{\geq 0}$: Number of direct depot-to-STS round-trips performed by shuttle k to STS i per shift.
- $x_{ijkm} \in \{0,1\}$:
  - 1 if sweeper k traverses arc (i,j) on trip m;
  - 0 otherwise.
- $q_{ikm} \geq 0$: Quantity of residual waste (tonnes) collected by sweeper k at node i during trip m.
- $T_{ikm} \geq 0$: Arrival time of sweeper k at node i during trip m (hours from shift start).
- $u_{ikm} \geq 1$: Auxiliary MTZ visit-order variable for sweeper k at node i on trip m.
- $N_k \in \mathbb{Z}_{\geq 0}$: Total number of physical vehicles of type k deployed.

3.1.4 Objective Function

$$\min Z = Z^{\text{Fuel}} + Z^{\text{Wages}} + Z^{\text{Carbon}} \tag{3.1}$$

$$Z^{\text{Fuel}} = \sum_{k \in K_S} \beta \; C_k^{\text{fuel}} \left[\sum_{i \in \mathcal{C}} 2 \; D_{0i} \, y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \beta \; C_k^{\text{fuel}} \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} \; x_{ijkm} \right] \tag{3.2}$$

$$Z^{\text{Wages}} = \sum_{k \in K_S \cup K_R} N_k \; C_k^{\text{wage}} \tag{3.3}$$

$$Z^{\text{Carbon}} = \sum_{k \in K_S} \lambda \; E_k \left[\sum_{i \in C} 2 \; D_{0i} \; y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \lambda \; E_k \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} \; x_{ijkm} \right] \tag{3.4}$$

The objective function Eq. [1] minimizes the total daily operational cost, composed of three components. Eq. [2] captures the distance-proportional fuel expenditure of both fleets, scaled by the congestion factor β. Eq. [3] represents the fixed daily labour cost incurred for each deployed vehicle, independent of distance travelled. Eq. [4] monetizes greenhouse gas emissions by applying the carbon penalty rate λ to the distance-based emission factors of both fleets.

3.1.5 Constraints

$$\sum_{k \in K_S} y_{ik} \; \geq \; 1, \qquad \forall \, i \in C \tag{3.5}$$

$$\sum_{k \in K_S} Q_k \; y_{ik} \; + \; \sum_{k \in K_R} \sum_{m \in M} q_{ikm} \; \geq \; \bar{S}_i + \Gamma \, \hat{S}_i, \qquad \forall \, i \in C \tag{3.6}$$

$$\sum_{j \in N} x_{ijkm} \; = \; \sum_{j \in N} x_{jikm}, \qquad \forall \, i \in N, \; k \in K_R, \; m \in M \tag{3.7}$$

$$\sum_{j \in C} x_{0jkm} \; \leq \; 1, \qquad \forall \, k \in K_R, \; m \in M \tag{3.8a}$$

$$\sum_{j \in N} x_{jikm} \;\leq\; 1, \qquad \forall\, i \in C,\; k \in K_R,\; m \in M \tag{3.8b}$$

$$q_{ikm} \;\leq\; Q_k \sum_{j \in N} x_{ijkm}\,, \qquad \forall\, i \in C,\; k \in K_R,\; m \in M \tag{3.9}$$

$$\sum_{i \in C} q_{ikm} \;\leq\; Q_k \sum_{j \in C} x_{0jkm}\,, \qquad \forall\, k \in K_R,\; m \in M \tag{3.10}$$

$$u_{ikm} - u_{jkm} + |C|\, x_{ijkm} \;\leq\; |C| - 1, \qquad \forall\, i,j \in C,\; i \neq j,\; k \in K_R,\; m \in M \tag{3.11}$$

$$E_i \sum_{j \in N} x_{ijkm} \;\leq\; T_{ikm} \;\leq\; L_i \sum_{j \in N} x_{ijkm}\,, \qquad \forall\, i \in C,\; k \in K_R,\; m \in M \tag{3.12}$$

$$T_{jkm} \;\geq\; T_{ikm} + \frac{D_{ij}}{v/\beta} + t_{\mathrm{LU}} - \mathcal{M}\left(1 - x_{ijkm}\right), \qquad \forall\, (i,j) \in A,\; j \neq 0,\; k \in K_R,\; m \in M \tag{3.13}$$

$$\mathcal{M} \;=\; T_{\max} \;+\; \max_{(i,j) \in A} \left[\frac{D_{ij}}{v/\beta} + t_{\mathrm{LU}}\right] \tag{3.14}$$

$$\sum_{m \in M} \sum_{(i,j) \in A} \left[\frac{D_{ij}}{v/\beta} + t_{\mathrm{LU}}\right] x_{ijkm} \;+\; B_{\mathrm{dur}} \;\leq\; T_{\max}, \qquad \forall\, k \in K_R \tag{3.15a}$$

$$\sum_{i \in C} y_{ik} \left[\frac{2D_{0i}}{v/\beta} + t_{\mathrm{LU}}\right] \;+\; B_{\mathrm{dur}} \;\leq\; T_{\max}, \qquad \forall\, k \in K_S \tag{3.15b}$$

$$N_k \geq \sum_{j \in C} x_{0jkm}, \qquad \forall k \in K_R,\ m \in M \tag{3.16}$$

$$N_k \geq \frac{1}{5} \sum_{i \in C} y_{ik}, \qquad \forall k \in K_S \tag{3.17}$$

Constraints (3.5) to (3.17) collectively define the operational feasibility, routing logic, and robustness of the proposed waste collection system. Constraint (3.5) guarantees that each secondary transfer station (STS) is served by at least one shuttle per shift, ensuring a minimum level of bulk waste extraction. Constraint (3.6) enforces robust demand satisfaction by requiring that the combined service capacity of shuttle and sweeper fleets meets or exceeds worst-case demand under uncertainty. Constraint (3.7) represents the classical flow-balance condition, ensuring route continuity by equating incoming and outgoing arcs for each vehicle, trip, and node. Constraint (3.8) restricts each sweeper to at most one departure from the depot per trip slot, preventing artificial route fragmentation, while Constraint (3.8b) enforces at most one visit per STS, ensuring structural consistency of routing decisions. Constraint (3.9) links service quantity to visitation, ensuring that waste is collected only when a node is visited, and zero otherwise. Constraint (3.10) limits total collected waste per trip to vehicle capacity, with automatic deactivation for unused trips. Constraint (3.11), the Miller–Tucker–Zemlin subtour elimination constraint[17], prevents infeasible cycles by enforcing a consistent visit ordering among nodes. Constraint (3.12) imposes hard time-window feasibility at each STS, activated only for visited nodes. Constraint (3.13) ensures temporal consistency by propagating arrival times along active arcs while accounting for travel and service times. Equation (3.14) defines a tight big-M constant to improve numerical stability and strengthen the linear relaxation. Constraint (3.15) limits total operational time per sweeper, including travel, service, and mandatory rest, ensuring shift feasibility. Constraint (3.15b) enforces an analogous shift limit for shuttle vehicles based on cumulative service time. Constraint (3.16) provides a lower bound on the number of deployed sweeper vehicles based on activated trip slots, ensuring sufficient fleet availability, while Constraint (3.17) establishes a similar lower bound for shuttle vehicles, reflecting their limited number of feasible round trips per shift.

3.2 Assumptions

To ensure computational tractability and model clarity, the following assumptions are adopted in this study:

- All Secondary Transfer Stations (STSs) are assumed to be fully operational throughout the planning horizon, with no disruptions or capacity limitations.
- All vehicles are assumed to have identical carbon emission rates, irrespective of their age, type, or operational condition.
- Fuel efficiency is considered uniform across the entire vehicle fleet, neglecting variations due to vehicle characteristics or maintenance conditions.
- Vehicle speed is assumed to be constant across all arcs within a given simulation scenario, thereby ignoring stochastic variations due to traffic dynamics or road conditions.
- The centralized landfill (depot) is assumed to have an unlimited availability of both vehicle types (i.e., shuttles and sweepers) as well as sufficient drivers, ensuring that any level of daily demand can be accommodated without resource constraints.

3.3 Solution Approach

The proposed solution adopts a hierarchical "cluster-first, route-second" methodology to efficiently address the computational complexity of large-scale vehicle routing problems. This hybrid framework integrates machine learning–based spatial decomposition with exact mathematical optimization, enabling the transformation of a large NP-hard problem into smaller, tractable sub-problems.

In the first stage, K-means clustering is applied to the geographical coordinates of the Secondary Transfer Stations (STSs) to partition the study area into spatially compact clusters. This decomposition reduces routing complexity by confining vehicle operations within localized zones and minimizing unnecessary inter-cluster travel. In the second stage, a heterogeneous fleet allocation strategy is employed within each cluster, where high-capacity

shuttle vehicles are assigned to handle bulk waste transport, while smaller sweepers are deployed to efficiently collect residual waste across multiple locations. This tiered allocation ensures better vehicle utilization and reduces operational inefficiencies associated with partial loads.

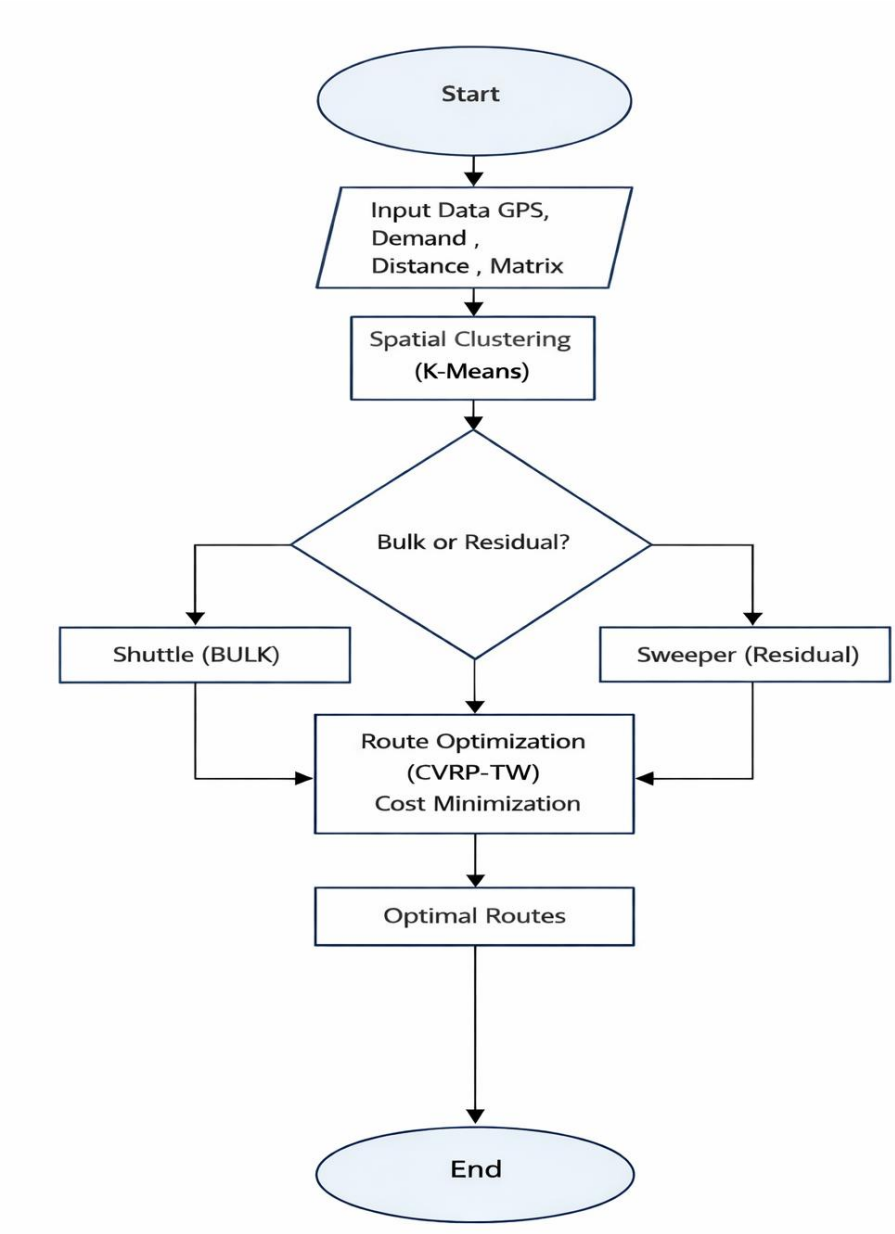


Figure 3.2: Flowchart of the overall process

Subsequently, a Mixed Integer Linear Programming (MILP) model is formulated and solved independently for each cluster to address the Capacitated Vehicle Routing Problem with Time Windows (CVRP-TW). The objective function minimizes a weighted combination of fuel costs, labor wages, and carbon emission penalties, while satisfying key operational constraints, including vehicle capacity limits, service time windows, and subtour elimination using the Miller–Tucker–Zemlin (MTZ) formulation[17]. Finally, the robustness of the proposed framework is evaluated through scenario-based stochastic simulations and sensitivity analyses under varying waste generation patterns, traffic congestion levels, and fuel price fluctuations. This comprehensive workflow ensures that the resulting routing strategies are not only cost-efficient but also resilient to real-world uncertainties.

CHAPTER 4

RESULT AND DISCUSSION

4.1 Data Collection

In this study, data were collected from Secondary Transfer Stations (STS) located across Rajshahi City to capture the spatial distribution and daily waste generation patterns.

A total of 20 STS locations were identified, covering major residential, commercial, institutional, and industrial areas within the city[18]. For each STS, the geographical coordinates (latitude and longitude) were recorded to facilitate spatial analysis and distance computation [20], which are essential inputs for the routing component of the proposed MILP model. In addition, the daily waste generation (in tons per day) at each STS was obtained to represent the demand that must be satisfied by the waste collection system. Data for this study were collected using both primary and secondary sources. Primary data were obtained from field observations, while secondary data were gathered from relevant existing records and literature.

Table 4.1 presents the collected dataset, including the names of STS locations, their corresponding coordinates, and the estimated waste generation per day. The waste generation values vary significantly across locations, reflecting the heterogeneity of land use and population density within Rajshahi City. For instance, high-demand areas such as RMC STS (31 tons/day), Mission Hospital STS (29 tons/day), and RU STS (28 tons/day) indicate zones with intensive activity, while relatively lower waste generation is observed in locations such as Haragram STS (8 tons/day) and Airport STS (11 tons/day).

Efficient waste collection requires the appropriate selection and deployment of vehicles with different capacities and operational characteristics. In this study, a heterogeneous fleet consisting of both sweeper vehicles (KR) and shuttle vehicles (KS) is considered to reflect real-world waste collection practices.

Table 4.2 summarizes the specifications of the vehicle types used in the model. Two types of sweepers with capacities of 1.5 tons and 3 tons are employed for localized waste collection within clusters, while a 5-ton shuttle vehicle is used for bulk transportation. The fuel cost varies by vehicle type, with higher-capacity vehicles incurring greater operational costs per kilometer. Similarly, driver wages increase with vehicle size, reflecting differences in operational requirements.

Emission factors are also incorporated to account for environmental impact. Larger vehicles produce higher emissions per kilometer, which directly influences the carbon cost component of the optimization model. The inclusion of heterogeneous vehicles allows the model to balance cost, capacity, and environmental considerations effectively.

Table 4.1: Waste collecting vehicle information

Vehicle Class	Role	Capacity (Qk Tons)	Fuel Cost (BDT/km)	Daily of driver Wage (BDT)	Emissions (kg/km)
Type 1 (1.5T)	Sweeper (KR)	1.5	18	1200	0.15
Type 2 (3.0T)	Sweeper (KR)	3	25	1500	0.22
Type 3 (5.0T)	Shuttle (KS)	5	35	1800	0.35

In addition to vehicle characteristics, several operational parameters are incorporated into the model to ensure realistic representation of waste collection activities. These parameters, presented in Table 4.3, capture traffic conditions, working hours, vehicle speed, and environmental costs.

The congestion factor (β) reflects traffic conditions in Rajshahi City and is used to adjust travel time and fuel consumption. The maximum working time (Tmax) is set to 8 hours, representing a standard daily labor shift. The average vehicle speed (v) ranges between 20 km/h, based on typical urban driving conditions for waste collection vehicles. The geographic coordinates of all nodes and the corresponding average traffic speeds were obtained from Google Maps [13]

Furthermore, a carbon penalty factor (λ) is included to quantify the environmental cost associated with vehicle emissions. This parameter enables the model to incorporate sustainability considerations by penalizing higher emissions in the objective function.

Table 4.2: Parameters involved

Parameter	Symbol	Value	Source/Description
Congestion Factor	β	1.3 - 1.5	Based on Rajshahi traffic audits
Max Working Time	Tmax	8 Hours	Standard labor shift
Average Speed	v	20 km/h	Typical city dumper speed
Carbon Penalty	λ	3.85[19]	BDT/kg of CO2 emissions

4.2 Result Analysis

4.2.1 Experiment of K-means clustering algorithm

The Secondary Transfer Stations (STS) were grouped using the **K-means clustering algorithm**. This method partitions STS locations into clusters based on their geographical coordinates (latitude and longitude), minimizing the distance within each cluster while maximizing separation between clusters.

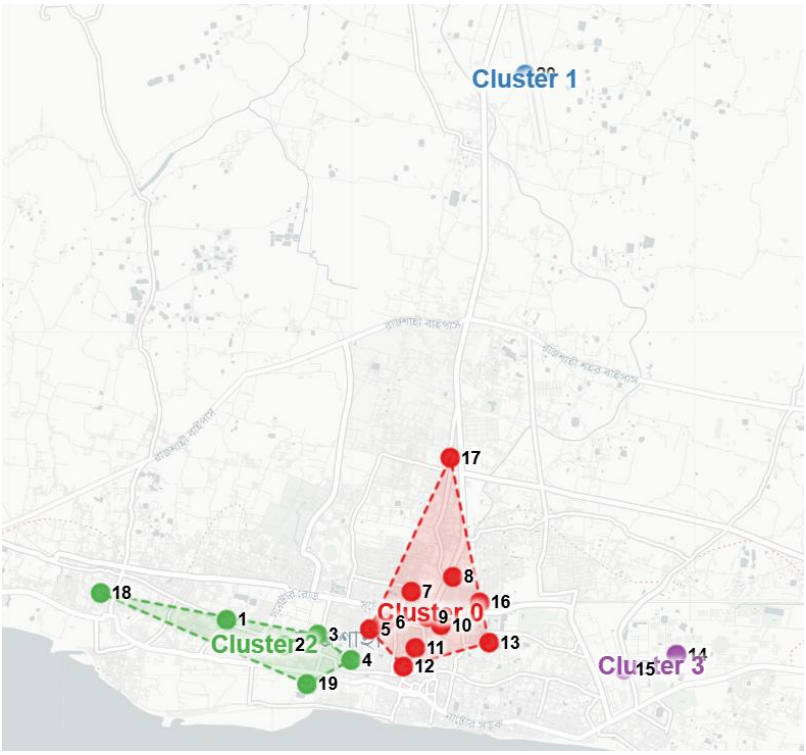


Figure 4.1: Clustering of STS

In this study, the STS nodes were divided into **four clusters (Cluster 0 to Cluster 3)**, as illustrated in Figure X. Each cluster represents a geographically compact service region, enabling more efficient route planning and reduced travel distances.

Table 4.3: STS names with their coordinates and waste generations of each STS per day

STS Name	Latitude	Longitude	Cluster ID	Waste Generated per Day (Ton)
Kashiadanga STS	24.3736	88.5778	2	9
Adorsho School STS	24.3705	88.5852	2	13
Court Walton STS	24.3718	88.5895	2	21
New Degree College STS	24.3689	88.5938	2	27
Mission Hospital STS	24.3725	88.5962	0	29
STS Bondhogate	24.3732	88.598	0	17
Terokhadia STS	24.3768	88.6015	0	13
Match Factory STS	24.3785	88.6068	0	24
STS near Sericulture	24.3738	88.6035	0	20
RMC STS	24.3729	88.6052	0	31
TTC STS	24.3702	88.602	0	14
STS Borokuthi	24.368	88.6005	0	18

Tikapara STS	24.3708	88.6115	0	25
Kazla STS	24.3695	88.6355	3	27
RU STS	24.3678	88.6288	3	28
Mansur Ali Park STS	24.3755	88.6102	0	20
Nowdapara STS	24.3925	88.6065	0	15
Haragram STS	24.3765	88.5616	2	8
Sopura STS	24.366	88.588	2	13
Airport STS	24.4372	88.616	1	11

4.2.2 Experiment of the proposed MILP model

To evaluate the effectiveness of the proposed MILP-based optimization model, a comparative analysis was conducted between the **traditional waste collection system** and the **optimized system obtained using Gurobi**. The results are summarized in Table 4.4 and illustrated in Figure 4.2.

Table 4.4 presents the total daily waste, traditional cost, optimized cost, and corresponding cost savings across multiple scenarios (Z_0–Z_{10}). The findings clearly demonstrate that the proposed optimization model consistently outperforms the traditional approach in all cases. The optimized costs are significantly lower, resulting in substantial economic benefits.

Table 4.4: Cost Reduction Analysis

Cost, Z_n	**Total Daily Waste, $\overline{S}_i$ (Tons)**	**Traditional Cost (BDT)**	**Optimized Cost (BDT)**	**Cost Saved (BDT)**	**Cost Reduction (%)**
0	459.6	119723.35	105223.67	14499.68	12.11
1	441.69	117160.22	96636.83	20523.39	17.52
2	470.6	119343.62	104943.62	14400	12.07
3	440.88	112451.35	98531.63	13919.73	12.38
4	441.41	115601.46	98114.24	17487.22	15.13
5	452.9	121108.08	101720.54	19387.54	16.01
6	470.52	121422.26	104902.52	16519.74	13.61
7	463.63	117123.26	104523.26	12600	10.76
8	452.16	113738.94	96815.05	16923.9	14.88
9	434.68	115911.26	100471.3	15439.95	13.32
10	439.72	120810.39	100616.72	20193.67	16.72

The **cost reduction ranges from 10.76% to 17.52%**, with an average reduction of approximately **13–15%** across all scenarios. The highest cost saving is observed in scenario **Z_1 (17.52%)**, while the lowest reduction occurs in **Z_7 (10.76%)**, indicating that the model maintains consistent performance even under varying waste generation conditions.

In terms of absolute savings, the model achieves reductions ranging from approximately **12,600 BDT to over 20,500 BDT per day**, which highlights the strong economic advantage of the optimized system. These savings are primarily attributed to **efficient route planning, better vehicle utilization, and reduced travel distances**, achieved through clustering and MILP-based routing.

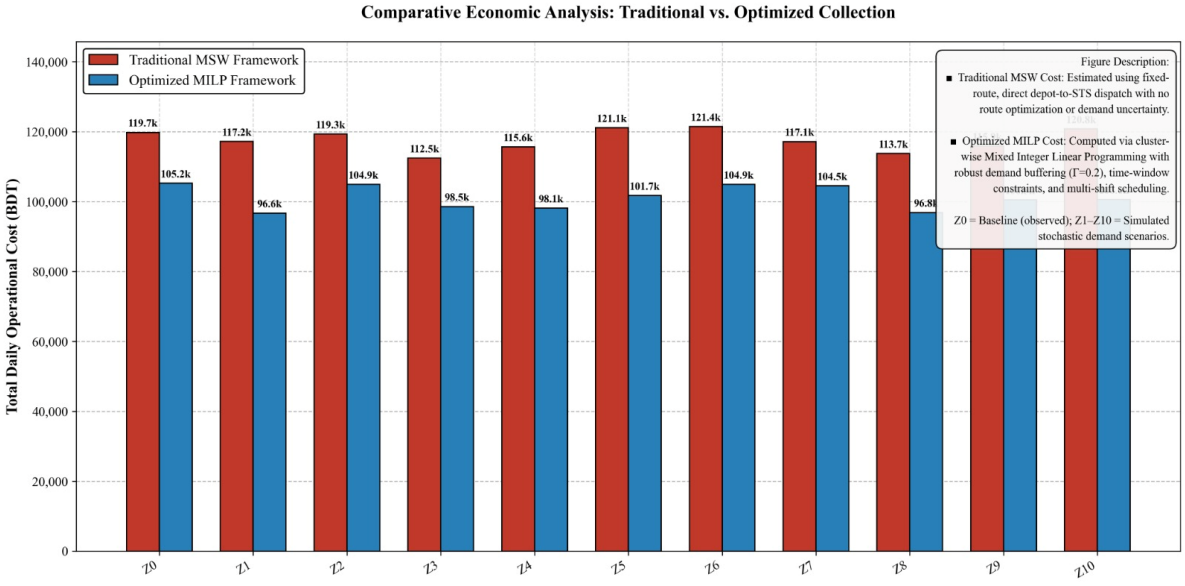


Figure 4.2: Comparative economic analysis: Traditional vs Optimal Collection

Additionally, the model demonstrates stability under varying demand levels, as the cost reduction remains relatively consistent despite fluctuations in total waste generation (ranging from approximately 434 to 470 tons per day). This indicates that the proposed approach is **scalable and adaptable** to different operational scenarios.

4.3 Sensitivity Analysis

To evaluate the robustness and responsiveness of the proposed optimization model, a sensitivity analysis was conducted by varying key parameters, including carbon tax (λ), fuel price, and traffic congestion factor (β).

4.3.1 Impact of Carbon Tax (λ)

The carbon penalty factor (λ) was varied from 0 to 8 BDT/kg CO_2 to assess the influence of environmental cost on the total objective function. The results indicate that increasing the carbon tax leads to a moderate increase in total cost, with a maximum rise of approximately 1.44% at the highest level.

Table 4.5: Impact of carbon tax

Carbon Tax λ (BDT/kg CO2)	Total Cost, Z (BDT)
0	158136.1
2	159247.9
3.85	160276.2
6	161471.4
8	162583.2

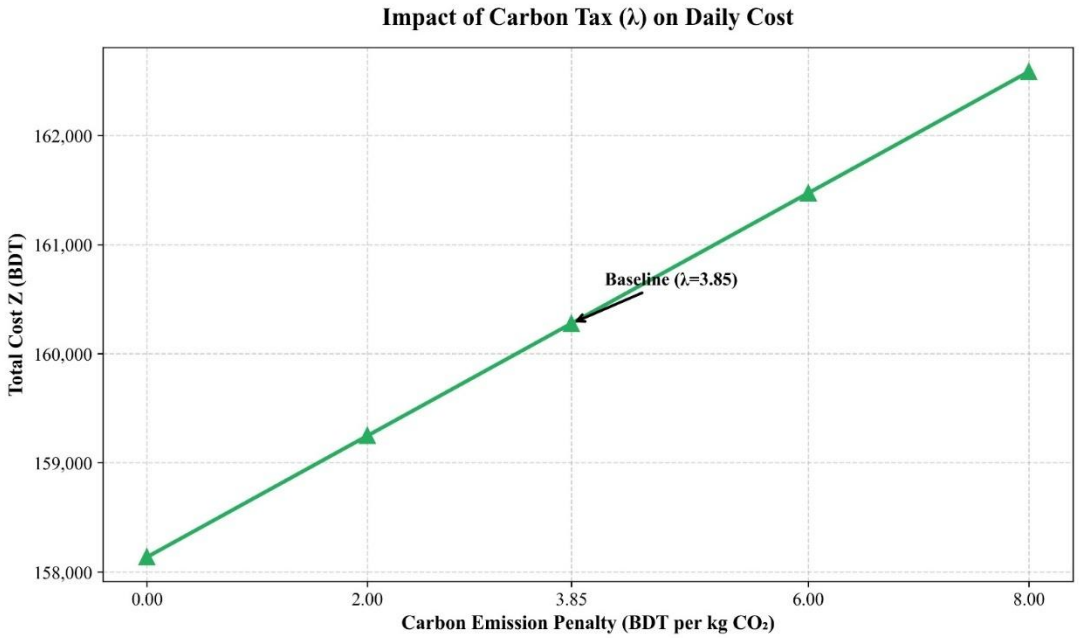


Figure 4.3: Impact of carbon tax on daily cost

At $\lambda = 0$, the total cost decreases by about 1.34%, reflecting the absence of environmental penalties. However, even at higher carbon tax levels, the overall cost variation remains relatively small. This suggests that the model is not highly sensitive to carbon pricing, primarily because fuel and operational costs dominate the objective function.

4.3.2 Impact of Fuel Price

Fuel price was varied from 80% to 120% of the baseline value to examine its effect on system cost. The results show a strong linear relationship between fuel price and total cost. A 20% decrease in fuel price reduces the total cost by approximately 9.89%, while a 20% increase results in a similar magnitude of cost increase.

Table 4.6: Impact of fuel price

Fuel Price Level	Multiplier	Total Cost Z (BDT)
80%	0.8	144429
90%	0.9	152352.6
Base (100%)	1	160276.2
110%	1.1	168199.9
120%	1.2	176123.5

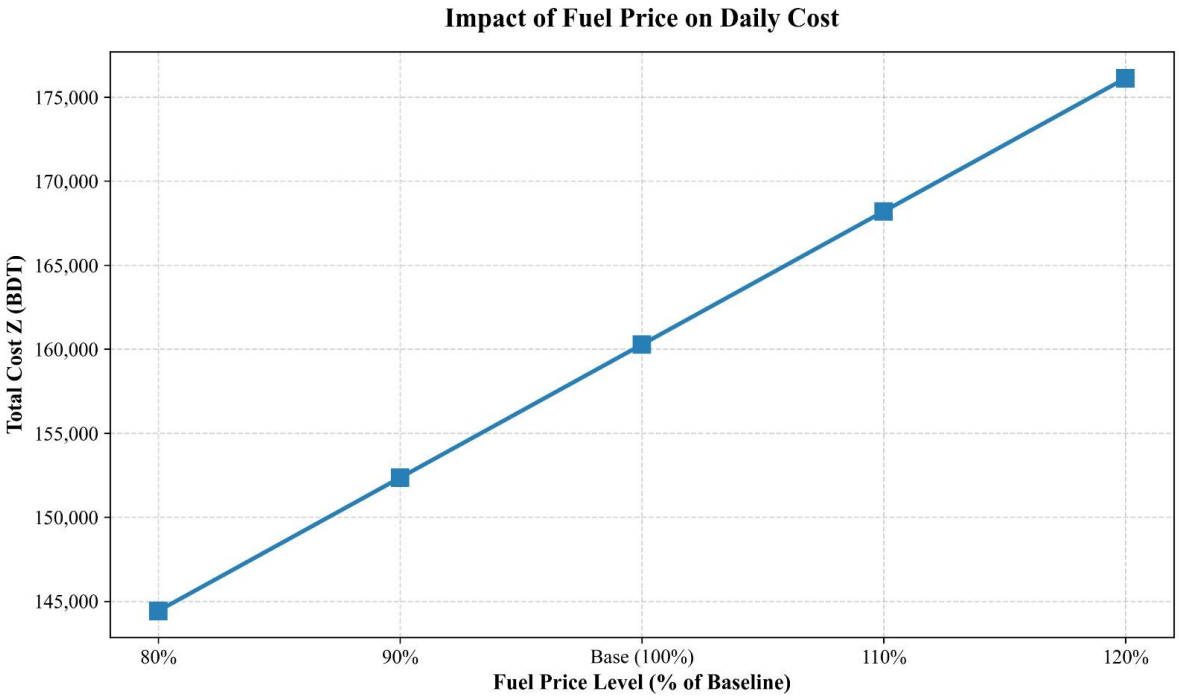


Figure 4.4: Impact of fuel price on daily cost

This indicates that fuel cost is one of the most influential parameters in the model, significantly affecting routing and operational decisions. Since fuel consumption is directly linked to travel distance and vehicle usage, even small variations in fuel price can lead to noticeable changes in total cost.

4.3.3 Impact of Traffic Congestion (β)

The traffic congestion factor (β) was varied between 1.0 and 1.8 to capture different traffic conditions in Rajshahi City. The results demonstrate that increasing congestion leads to a substantial rise in total cost, with an increase of approximately 18.62% at β = 1.8.

Table 4.7: Impact of traffic congestion

Traffic Factor β	Total Cost Z (BDT)	Traffic Factor β
1	132237.36	1
1.2	147156.8	1.2
1.4	160276.24	1.4
1.6	175195.68	1.6
1.8	190115.12	1.8

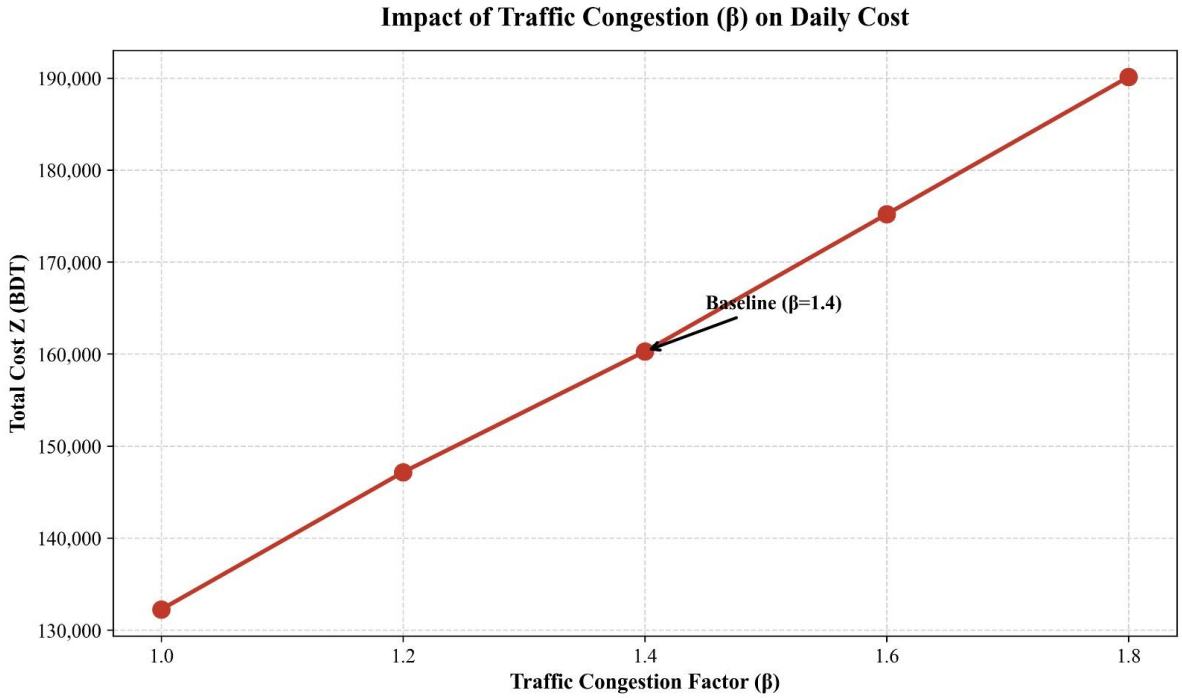


Figure 4.5: Impact of traffic congestion on daily cost

Conversely, under low congestion conditions (β = 1.0), the total cost decreases by around 17.49%**.** This highlights that the model is highly sensitive to traffic conditions, as congestion directly affects travel time, fuel consumption, and operational efficiency.

4.3.4 Combined Effect of Fuel Price and Traffic Congestion

A combined sensitivity analysis was conducted using varying levels of both fuel price and traffic congestion, as shown in the heatmap results. The analysis reveals that the interaction between these two parameters has a significant compounded effect on total cost.

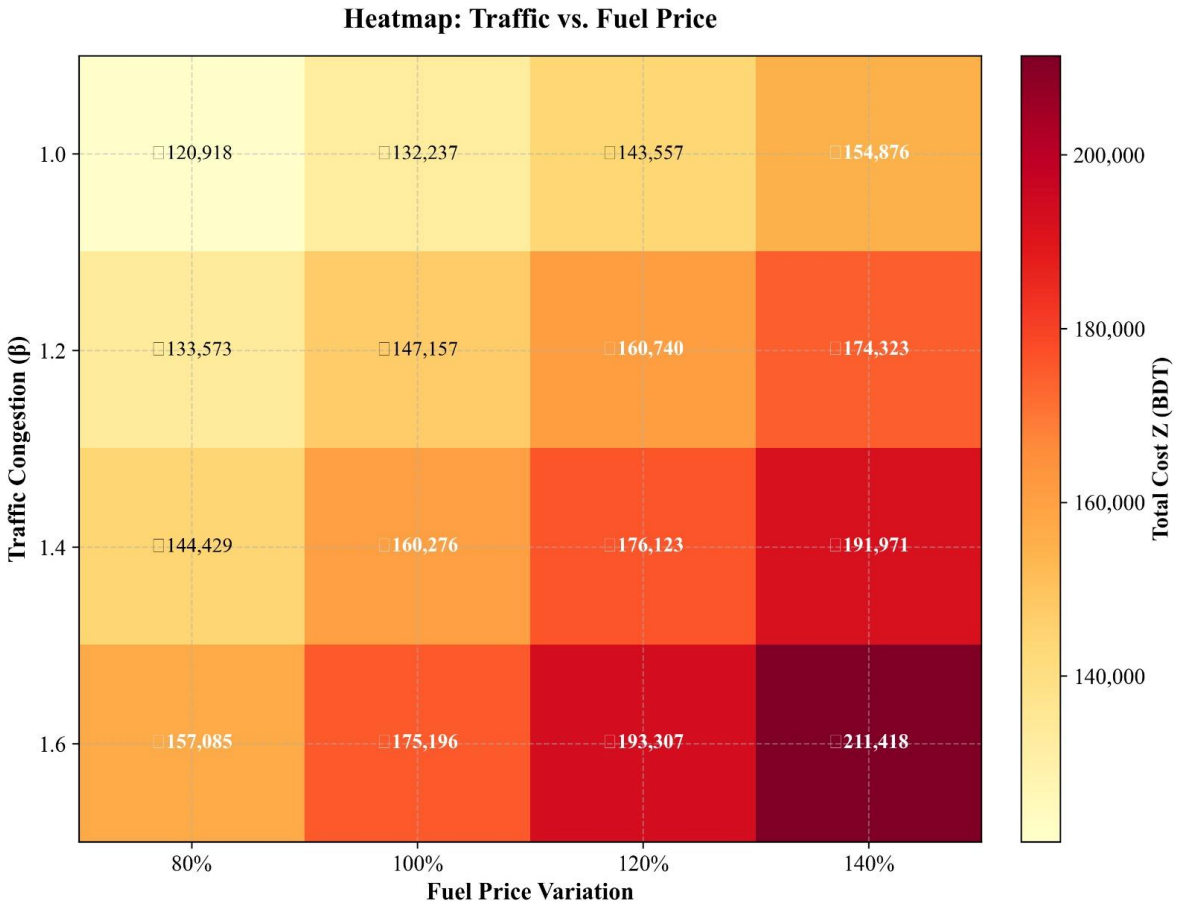


Figure 4.6: Heatmap of traffic vs fuel price on daily cost

For instance, at low congestion (β = 1.0) and reduced fuel price (80%), the total cost decreases by approximately 24.56%, representing the most favorable scenario. In contrast, higher fuel prices combined with increased congestion lead to substantial cost increases.

These results confirm that fuel price and traffic congestion are the dominant factors influencing system performance, and their combined impact is more significant than individual effects.

4.4 Discussion

4.4.1 Model Performance

The proposed K-Means–MILP framework demonstrates significant improvement over conventional routing methods, achieving cost reductions of 12.07%–17.52% across all scenarios. In the baseline case (Z_o = 459.6 tons), daily cost is reduced from 119,723 BDT to 105,223 BDT, saving approximately 14,500 BDT.

These gains result from eliminating spatial inefficiencies. K-Means clustering localizes operations, while heterogeneous fleet allocation ensures efficient handling of bulk and residual waste. The MILP model further enforces capacity (Qk) and shift constraints ($Tmax$), minimizing redundant travel and underutilization.

4.4.2 Sensitivity to Urban Dynamics

1) Traffic Congestion (β)

The system is highly sensitive to congestion. Increasing β from 1.0 to 1.8 results in cost variation from −17.49% to +18.62%. Higher travel times necessitate additional vehicles to satisfy time constraints, significantly increasing operational cost. This highlights the importance of off-peak scheduling.

2) Fuel Price (C_{fuel})

A ±20% change in fuel price leads to a moderate cost variation of ±9.89%, indicating that optimized routing reduces fuel dependency and improves economic resilience.

4.4.3 Environmental Policy Impact

Introducing a carbon penalty (λ)

(λ) produces a non-linear response. While total cost increases marginally (up to 1.44%), the model exhibits fleet-shifting behavior, favoring more emission-efficient vehicles. This confirms the framework's effectiveness in supporting sustainable logistics policies.

4.4.4 Combined Stress Effects

Two-way sensitivity analysis (traffic and fuel) reveals non-linear cost escalation under compounded conditions. Costs decrease by 24.56% under optimal conditions but rise sharply under high congestion and fuel prices. This demonstrates the need for dynamic, real-time planning rather than static budgeting.

4.4.5 Practical Implications

The framework enables data-driven routing for the Rajshahi City Corporation (RCC), improving cost efficiency, operational feasibility, and environmental compliance. Its scalable architecture makes it suitable for broader application in rapidly urbanizing regions.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The rapid growth of urban populations has rendered traditional, heuristically managed waste collection systems both economically inefficient and environmentally unsustainable. In response, this study developed and computationally validated an integrated decision-support framework for the Rajshahi City Corporation, combining K-means spatial clustering with a comprehensive Mixed Integer Linear Programming (MILP) routing model. The proposed framework effectively minimizes a tripartite objective function encompassing fuel costs, labor wages, and carbon emission penalties. By rigorously incorporating operational constraints including heterogeneous vehicle capacities, mandatory subtour elimination, and strict labor shift time windows the model successfully transforms the conventional radial dispatch system into an optimized multi-trip routing network. Comparative scenario analysis (Z0 through Z10) demonstrates that the integration of machine learning–based spatial zoning with exact optimization significantly reduces fleet size requirements and eliminates redundant travel distances, resulting in substantial cost savings.

5.2 Recommendations

While the proposed model demonstrates strong performance and practical applicability, several extensions remain for future research. In the current cluster-first route-second framework, clustering is performed solely based on distance. However, future studies can enhance the approach by incorporating both distance and capacity considerations to improve cluster balance and overall routing efficiency. Incorporating stochastic or fuzzy MILP formulations would enable more accurate representation of demand uncertainty in waste generation Additionally, extending the framework to a multi-commodity flow model would support reverse logistics and source-level waste segregation, aligning with circular economy

objectives. To address scalability challenges in large urban environments, hybrid optimization approaches combining MILP with metaheuristic techniques such as Variable Neighborhood Search or Simulated Annealing should be explored. Furthermore, integrating real-time Geographic Information Systems (GIS) and live traffic data would allow dynamic modeling of congestion parameters, enabling adaptive and real-time routing decisions. Finally, future models should incorporate vehicle-specific characteristics, such as varying fuel efficiency and carbon emission rates based on vehicle age, type, and capacity, to better reflect real-world fleet heterogeneity and improve the accuracy of cost and environmental impact assessments. Overall, this research provides municipal authorities with a robust and deployable mathematical tool for achieving efficient and sustainable urban waste management.

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